

Xiaoyang Chen

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Career Profile

- Data-driven IT professional specialising in **data analysis, data engineering, and artificial intelligence** in UNSW course project.
- Experienced in **SQL query optimisation, ETL pipeline development, and cloud-based analytics** using Snowflake and BigQuery in internship of Zhongfutong Group, Fuzhou, China.
- Proficient in **Python (Pandas, NumPy)** for data cleaning and transformation, with strong skills in data visualisation and dashboard creation in UNSW course project.

Education

- **University of New South Wales (UNSW)**, Sydney, Australia | **WAM: 68/100** Sep 2024 – Aug 2026 (Expected)
Master of Information Technology (*Data Science and Engineering*)
Relevant Courses: Machine Learning, Data Mining, AI, Statistics, Python
- **Xiamen University of Technology**, Xiamen, China Sep 2020 – Jul 2024
Bachelor of Engineering (*Intelligent Manufacturing Engineering*)
Third Prize Scholarship (2023) | Outstanding Graduate (2024)
- **Xiamen University of Technology**, Xiamen, China Mar 2021 – Jun 2024
Bachelor of Engineering (*Computer Science and Technology*)

Relevant Experience

- IT Intern | Zhongfutong Group, Fuzhou, China** Jul 2023
 - Optimised internal data pipelines by automating daily reporting scripts and **MySQL** queries, reducing runtimes by **30%**
 - Integrated **OpenIM** for internal messaging and built delivery monitoring logic
 - Managed **Node.js** front-end and back-end deployment (Windows & Ubuntu) to ensure pipeline stability

Relevant Project

- AI Forecasting Project | Solar Radiation Prediction with Neural Networks** Mar 2026
 - Developed a **PyTorch-based neural network model** to forecast solar radiation from time-series data.
 - Engineered Fourier cyclical features, 7-day rolling statistics, and lagged variables.
 - Built and compared single-output and multi-output models for 7-day-ahead forecasting.
 - Applied **MC Dropout and attention heatmaps** to improve uncertainty analysis and interpretability.
 - Validated performance through **ablation studies, paired t-tests, and structured error analysis**
- Database Systems Course Project | UNSW Sydney, Australia** Apr 2025
 - Led an analysis of **59,000+ NSW fuel price records** to identify regional pricing patterns and improve dataset reliability
 - Applied **Pandas and NumPy** and built a postcode-suburb algorithm, increasing data accuracy **from ~50% to ~66%**
 - Created **PriceChangeAverage** metric to highlight anomalies, revealing unusual pricing trends in Western Sydney
- Database Systems Course Project | UNSW Sydney, Australia** Oct 2024
 - Developed a **PostgreSQL-based reporting system** for large academic datasets to improve data extraction and analysis
 - Optimised SQL queries and implemented **PL/pgSQL** functions, reducing database load and enhancing reporting speed
 - Automated workflows, facilitating accessibility for analytical insights and earning a **99/100 project score**

Volunteer Activities

- Volunteer - International Collegiate Programming Contest (ICPC) Preliminary | UNSW** Sep 2025
 - Managed one examination room, ensuring smooth contest operations and participant coordination
 - Set up and maintained the **competition system environment** (hardware and software) before the contest

- Supervised test conduct, handled real-time technical issues, and ensured rule compliance
 - Performed **post-exam maintenance and data collection**, ensuring accurate and secure submission handling
- Leadership Foundations Program(AHEGS-recognised UNSW program) | UNSW Employability** Apr 2025
- Developed leadership, communication, and teamwork skills
 - Enhanced self-awareness, resilience, and inclusivity in group settings
 - Gained experience in building effective and collaborative teams
- Volunteer - 21st China International Fair for Investment and Trade (CIFIT) | Xiamen, China** Mar 2021
- Assisted with cost monitoring and financial tracking at a major trade fair; awarded Volunteer Certificate

Honours and Awards

- 2nd Prize - 6th National Mathematics Competition for College Students (Non-mathematics Group) - National Level Jul 2023
- 1st Prize - 2023 Algorithm Competition for College Students - National Level Apr 2023
- 3rd Prize - 4th National Mathematical Ability Challenge - National Level Mar 2023

Technology Skills

- **Programming and Scripting:** Python (Pandas, NumPy), R, SQL, Java, C, C++, JavaScript, HTML
- **Data Engineering and Databases:** MySQL, PostgreSQL, SQL (JOINS, Views, Group By), ETL pipelines, Git, Linux
- **Data Analysis and Visualisation:** Excel, Google Sheets, MATLAB, Visualisation Dashboards, Matplotlib
- **Machine Learning and Big Data Tools:** Decision Trees, K-means, DBSCAN, Naïve Bayes, Cross-validation, BigQuery

Certifications

- Software Copyright Registration - "Analytic Hierarchy Process Data Evaluation Software V1.0" Jan 2025
- CS50x: Introduction to Computer Science | Harvard University Nov 2024
- Certificate of Occupational Ability of Business Data Analysts (Level 4) Aug 2024